

Transformer tests

Short-Circuit Test

$$Z_{eq} = R_{eq} + jX_{eq}$$

$$Z_b = \frac{V_n^2}{S_b}, \quad I_n = \frac{S_b}{\sqrt{3}V_n}$$

$$V_{sc} \frac{V_n}{\sqrt{3}} = Z_{eq} I_n$$

$$V_{sc} = \frac{Z_{eq}}{Z_b} = Z_{eq}(p.u.)$$

$$V_{sc}(\%) = Z_{eq}(p.u.)$$

$$P_{sc} = 3R_{eq}I_n^2$$

$$P_{sc}(p.u.) = \frac{3R_{eq}I_n^2}{S_b} = \frac{R_{eq}}{Z_b} = R_{eq}(p.u.)$$

$$P_{sc}(p.u.) = R_{eq}(p.u.)$$

Open-Circuit Test

$$Y_0 = G_{fe} + jB_m$$

$$Y_b = \frac{1}{Z_b}$$

$$I_0 I_n = \frac{V_n}{\sqrt{3}} Y_0$$

$$I_0 = \frac{Y_0}{1/Z_b} = \frac{Y_0}{Y_b}$$

$$I_0(\%) = Y_0(p.u.)$$

$$P_0 = 3 \left(\frac{V_n^2}{\sqrt{3}} \right)^2 G_{fe}$$

$$P_0(p.u.) = \frac{V_n^2 G_{fe}}{S_b} = \frac{G_{fe}}{1/Z_b} = G_{fe}(p.u.)$$

$$P_0(p.u.) = G_{fe}(p.u.)$$